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MetaCommunityMetrics.jl: A Julia Package for  
Spatiotemporal Metacommunity Analysis

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Yan Yin Jenny Cheung <sup>1</sup> and Laura Melissa Guzman <sup>1</sup>¶

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1 Department of Entomology, Cornell University, United States ¶ Corresponding author

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Summary

6 Community ecologists are often interested in inferring ecological processes from observational  
7 data. This involves calculating different metrics such as beta diversity, which measures the  
8 dissimilarity between two communities in terms of species richness or abundance. Calculating  
9 these metrics is easy and fast due to the availability of R packages, particularly for datasets with  
10 a limited number of sites and sampling events. However, community ecology today is growing  
11 beyond small-scale, single-dataset analyses. Large biodiversity databases have expanded the  
12 sets of spatiotemporal community data, and there is a growing interest in using simulations to  
13 study ecological processes. While R packages can achieve high performance through tools such  
14 as Rcpp, this requires re-implementation in C++, which presents a barrier for most ecologists.  
15 Julia offers native high performance while remaining as accessible as R, making it a more  
16 practical path to efficient computation for large-scale ecological analyses. However, broader  
17 adoption of Julia in ecology depends on having a rich ecosystem of packages, which this  
18 package aims to contribute to.

19 MetaCommunityMetrics.jl is a Julia package that helps ecologists analyze datasets of  
20 species observations to understand how community composition varies across space and  
21 time. By implementing computationally optimized versions of established software and new  
22 complementary functions designed for large spatiotemporal datasets, this package enables  
23 ecologists to efficiently perform analyses that would be computationally prohibitive with existing  
24 tools. We designed intuitive interfaces that would feel familiar to ecologists transitioning from  
25 R. Our function naming conventions closely match their R equivalents, and we maintained  
26 parameter naming and ordering consistency where possible. Some of the metrics we have  
27 implemented include beta diversity decompositions in space and time, dispersal-niche continuum  
28 index, and variability metrics measuring temporal variability in abundance across spatial scales  
29 and organizational levels of a metacommunity. A full list is provided in [Table 1](#). Full  
30 documentation and examples, including installation instructions and help files, are available at  
31 <https://cralibe.github.io/MetaCommunityMetrics.jl/>.

**Table 1:** A summary of the metrics and functions available in MetaCommunityMetrics.jl.

| Metrics                                         | Functions          | Details                                                                        | Ecological Meaning                                                                                                |
|-------------------------------------------------|--------------------|--------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------|
| Beta diversity decompositions in space and time | beta_diversity()   | Re-implementation of R package adespatial( <a href="#">Dray et al., 2023</a> ) | Community dissimilarity decomposed into species replacement and richness/abundance differences                    |
|                                                 | spatial_beta_div() | Re-implementation of R script from Guzman et al. ( <a href="#">2022</a> )      | Spatial variation in community composition decomposed into species replacement and richness/abundance differences |

| Metrics                                | Functions                    | Details                                                             | Ecological Meaning                                                                                                 |
|----------------------------------------|------------------------------|---------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------|
| Dispersal-niche continuum index (DNCI) | temporal_beta_div()          | Re-implementation of R script from Guzman et al. (2022)             | Temporal variation in community composition decomposed into species replacement and richness/abundance differences |
|                                        | DNCI_create_groups()         | A novel grouping function for DNCI analysis                         | Groups sites geographically using hierarchical clustering for DNCI analysis                                        |
|                                        | DNCI_plot_groups()           | A novel visualization function for DNCI analysis                    | Visualizes geographic site groupings for DNCI analysis                                                             |
|                                        | DNCI_multigroup()            | Re-implementation of R script from R package DNCImer (Gibert, 2024) | Identifies whether dispersal or niche processes drive community assembly                                           |
| Niche overlap index                    | niche_overlap()              | Novel implementation                                                | Mean niche overlap between all species pairs based on relative abundance distributions                             |
| Occupied patches proportion            | prop_patches()               | Re-implementation of R script from Guzman et al. (2022)             | Proportion of sites occupied by each species                                                                       |
| Variability Metric                     | CV_meta()                    | Re-implementation of R script from Wang et al. (2019)               | Temporal variability in abundance across spatial scales and organizational levels                                  |
| Niche hypervolume measurements         | MVNH_det()                   | Re-implementation from R package MVNH(Lu et al., 2021)              | Volume of a species' niche hypervolume                                                                             |
|                                        | MVNH_dissimilarity()         | Re-implementation from R package MVNH(Lu et al., 2021)              | Niche dissimilarity between two species using Bhattacharyya distance                                               |
|                                        | average_MVNH_det()           | Novel extension of MVNH_det()                                       | Mean niche hypervolume across all species in a community                                                           |
|                                        | average_MVNH_dissimilarity() | Novel extension of MVNH_dissimilarity()                             | Mean niche dissimilarity across all species pairs in a community                                                   |

Statement of need

Many valuable R (R Core Team, 2024) packages have been developed to aid in community analyses by providing ecological metrics that can summarize processes from biodiversity patterns and have been widely adopted in the field, such as adespatial (Dray et al., 2023), codyn (Hallett et al., 2020), and vegan (Oksanen et al., 2024). However, conducting community analyses in R is often computationally expensive, especially when working with large datasets, a limitation that becomes critical in workflows requiring repeated metric calculations. These include simulation studies for method validation, bootstrap resampling for uncertainty estimation

(Efron, 1992), null model approaches for statistical testing (Gotelli & Graves, 1996), and the analysis of large biodiversity databases, such as BioTime (Dornelas et al., 2025), all of which can create significant computational bottlenecks when processing multiple large datasets iteratively. To improve the efficiency of these computationally intensive analyses, Julia becomes a logical choice for re-implementations. Julia is known for its just-in-time compilation that optimizes code execution to levels comparable with lower-level languages like C or Fortran (Bezanson et al., 2017), yet Julia remains relatively uncommon in ecological workflows.

## State of the field

To position `MetaCommunityMetrics.jl` within the Julia ecosystem, we conducted a comprehensive review of existing Julia packages under the `EcoJulia`, a community-driven effort providing tools for ecological and environmental analysis, and the `PoisotLab`, which develops many Julia packages focusing on quantitative and computational ecology. The packages under `EcoJulia` cover a range of ecological analyses, including spatial ecology, diversity measurement, phylogenetics, and ecosystem simulations, while the `PoisotLab` packages focus primarily on species distributions and species interaction networks. While some of these packages implement diversity measures relevant to community ecology, none implement the specific metacommunity-scale metrics listed in Table 1. We ensured that our package builds beyond the tools currently existing in Julia even when sharing similar ecological concepts. For example, our beta diversity metrics share the concept of beta diversity with `Diversity.jl` (Robert Reeve & Harris, 2024). However, our implementation decomposes beta diversity into species replacement and richness differences (or abundance difference when abundance data is used instead of occurrence data) (Baselga, 2010, 2012, 2013; Legendre, 2014), which differentiates our approach from `Diversity.jl`, which emphasizes alpha, beta, and gamma diversity measures at the metacommunity and community levels (Richard Reeve et al., 2014). `MetaCommunityMetrics.jl` is, to our knowledge, the first Julia package to implement the metrics listed in Table 1, which are essential for research in community ecology but were previously only available through computationally expensive R implementations. However, some of the `EcoJulia` and `PoisotLab` packages are compatible with `MetaCommunityMetrics.jl` as they can be used upstream to retrieve and process data. For instance, `SpeciesDistributionToolkit.jl` provides a wrapper for the GBIF API (GBIF: The Global Biodiversity Information Facility, 2024) that allows users to query and retrieve georeferenced species occurrence records, which can then be formatted into a site-by-species matrix and analyzed using `MetaCommunityMetrics.jl` to investigate metacommunity processes, such as the interplay between dispersal and niche-based processes in shaping community composition across space and time.

## Software Design

We built `MetaCommunityMetrics.jl` on three design principles: usability, computational efficiency, and accuracy.

First, we prioritized usability for spatiotemporal data and simulation workflows. We extended single-snapshot metrics (single site and/or time point) to operate on multidimensional datasets (multiple sites and/or time points) via aggregation functions (`spatial_beta_div()`, `temporal_beta_div()`, `average_MVNH_det()`, `average_MVNH_dissimilarity()`), eliminating manual looping. Our DNCI implementation also accepts empty sites and singletons using Zero-adjusted Bray-Curtis dissimilarity, filters uninformative species (completely absent or ubiquitous), and implements status flags for five edge cases rather than throwing errors or returning NAs, supporting simulation workflows where sparse matrices are meaningful. We developed `DNCI_create_groups()` and `DNCI_plot_groups()` to automate site grouping (required by Vilmi et al. (2021) but manual in R) and visualize grouping results. See

89 [documentation](#) for details.

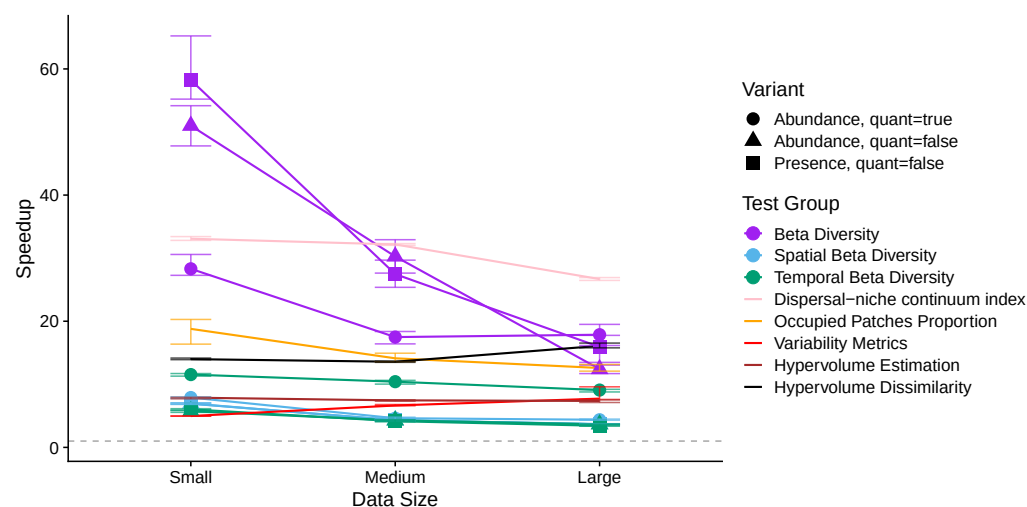
90 Second, we prioritized computational efficiency over comprehensive coverage (compared to the  
91 R implementations). Our beta diversity focuses on the Podani family approach (a mathematical  
92 framework that decomposes pairwise dissimilarity between sites into two distinct ecological  
93 components: species replacement and richness differences) rather than all five coefficient  
94 families, `MVNH_dissimilarity()` implements only the Bhattacharyya distance (a measure of  
95 dissimilarity between two probability distributions used here to quantify niche dissimilarity  
96 between two multivariate normal hypervolumes), and DNCI lacks built-in visualizations. We plan  
97 to expand these in future releases. To maximize performance, we use vectorized operations where  
98 possible and employ Cholesky decomposition in `MVNH_dissimilarity()` for the Mahalanobis  
99 component of the Bhattacharyya distance calculation, replacing the LU decomposition (a matrix  
100 factorization method that expresses a matrix as the product of a lower and upper triangular  
101 matrix) used in `solve()` in the original R implementation, providing substantial performance  
102 gains for positive-definite matrices ([Haddad, 2024](#)).

103 Finally, to ensure accuracy, all implementations with equivalent R functions were validated using  
104 the package's built-in sample data by running both Julia and R implementations on identical  
105 inputs within the same script via `RCall`, and computing the absolute differences between outputs.  
106 For all deterministic functions, differences were smaller than 0.0001, which we considered  
107 negligible and attributable to floating-point differences between implementations. For the  
108 DNCI, which is permutation-based, validation was conducted by running both implementations  
109 100 times with 1000 permutations each and comparing the distributions of results, which  
110 showed agreement in ecological interpretation across all group pairs (see [documentation](#)).

## 111 Research Impact Statement

112 `MetaCommunityMetrics.jl` delivers substantial computational advantages over existing tools.  
113 For example, we re-implemented the `beta.div.comp` function from the R package `adespatial`  
114 ([Dray et al., 2023](#)), achieving up to  $\sim 51\times$  faster execution times while maintaining  
115 methodological consistency (see [Figure 1](#)). As shown in [Figure 1](#), performance gains range  
116 from  $\sim 3\times$  to  $51\times$  across all re-implemented functions using [three testing datasets](#) (large:  
117 53,352 observations; medium: 26,676 observations; small: 5,325 observations), positioning  
118 `MetaCommunityMetrics.jl` as an efficient solution for diverse ecological research needs, from  
119 standard biodiversity analyses to computationally intensive simulation studies and large-scale  
120 data processing. Memory usage is generally comparable between Julia and R implementations,  
121 with notable reductions in Julia for the variability metrics function (e.g., 12.60 MiB vs 60.23  
122 MiB for the large dataset). However, the DNCI implementation requires substantially more  
123 memory in Julia than in R (e.g., 803.55 MiB vs 77.79 MiB for the large dataset). This  
124 difference arises because the R implementation generates the column-sum-preserving permuted  
125 matrices one at a time within the permutation loop, immediately computing the result and  
126 discarding the matrix, so only a single permuted matrix is held in memory at any given time.  
127 The use of parallelization in the R benchmark further reduces the peak memory usage. In  
128 contrast, the Julia implementation generates all permuted matrices upfront and stores them  
129 simultaneously in memory before processing. Direct comparisons should also be interpreted  
130 with caution given the differences in how each language measures memory allocation (see  
131 [documentation](#) for details).

132 Beyond computational performance, the development of `MetaCommunityMetrics.jl` was driven  
133 in part by the needs of an ongoing empirical application of Approximate Bayesian Computation  
134 methods to connect metacommunity theory with observed data from a long-term phytoplankton  
135 community monitoring dataset. Specifically, the requirement to efficiently calculate summary  
136 statistics for this application directly informed the design and implementation of several  
137 functions in the package. The manuscript of this application is under preparation.



**Figure 1:** A plot showing the speedup of all benchmarked functions across three datasets: small (5,335 observations, 10% of the full dataset), medium (26,676 observations, 50% of the full dataset), and large (53,352 observations), all derived from the sample dataset available via `load_sample_data()`. The sample dataset is a subset of rodent abundance data from the Portal Project (Ernest et al., 2018), a long-term study of a Chihuahuan desert ecosystem, covering 24 sites, 21 species, and 117 sampling events from 2010 to 2023. The small and medium datasets were generated by random subsampling of the sample dataset with a fixed random seed and saved for consistent reuse across benchmark runs. Each function was benchmarked using 100 samples in both Julia and R. Speedup is calculated as the R median execution time divided by the Julia median execution time. Median speedup and its 95% bootstrap confidence interval (based on 2000 bootstrap iterations) are shown for each comparison. The grey dashed line indicates where speedup equals 1, meaning both implementations require the same execution time. Beta diversity has two variants: `quant = true` (abundance data) and `quant = false` (occurrence data). When `quant = false`, abundance and presence inputs produce equivalent results, but the abundance input requires an additional internal conversion to binary, which accounts for the difference in speedup between the two inputs. Details can be found in the [documentation](#).

## AI usage disclosure

We used Claude, a generative AI tool, for proofreading and identifying grammatical errors in the manuscript and software documentation, and for providing optimization suggestions during software development. All writing suggestions were manually reviewed by the authors and selectively accepted. For optimization suggestions during software development, all suggestions were manually reviewed by the authors and selectively accepted, verified to produce the same results as before optimization, and benchmarked to confirm improvements in execution time and/or memory usage.

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